UNIVERSITY OF CALGARY

Graduate Award Competition

UCID Number:	30210220		
Application Number:	10027229	**Award Year:**	2024
Last Name:	Droobi	**First Name:**	Ahmad

Current Program

Program currently registered in Graduate Studies Master's Thes

Degree Master of Science

Area of study Mechanical Engineering

Start Term of Degree 2237

Registration Status Full-Time

Program you will request funding for

Start term of degree Fall 2023

Graduate Program Graduate Studies Master's Thes

Degree Master of Science

Area of study Mechanical & Manufacturing Engineering

Competition Selection

Alberta Innovates Graduate Student Scholarships

***Alberta Innovates Degree Stream**

Master's

***Detail how you will engage with industry during your graduate studies (254 char limit)**

I plan to engage industry through internships, joint projects, and knowledge exchange sessions. Seeking partnerships for data sharing, validation, and real-world application of research findings, collaborations beneficial for both academia and industry.

***Detail your involvement with industry, indicate if you are not currently involved with industry (254 char limit)**

As an academic researcher, I collaborate with industry through consultations, sharing insights, and presenting findings in workshops or conferences. Currently not directly engaged in industry, but open to partnerships for practical applications

***Detail how your research aligns with the Alberta Innovates Innovation Priority Area selected (254 char limit)**

By merging particle tracing and predictive machine learning, our research significantly enhances air quality prediction, weather forecasting, and informed decision-making, pivotal for environmental sustainability.

***Detail how your research can produce potential environmental, health, economic, social and other impacts in Alberta (254 char limit)**

Our research can revolutionize air quality management, aiding health by reducing pollution exposure. Accurate weather forecasts benefit agriculture, economy, and disaster preparedness, impacting Alberta's environmental sustainability and societal being.

***Detail how your research aligns with the Alberta Innovates Platform/Emerging Technology Area selected (254 char limit)**

The research aligns with Alberta Innovates' ICT Emerging Technology Area by integrating particle tracing and predictive machine learning to advance environmental forecasting, leveraging AI/ML for data analytics and modeling, contributing to Advanced Data

***Alberta Innovates Innovation Priority Area**

Environment

***Alberta Innovates Emerging Technology Area**

ICT Advanced Data Management and Analytics including Artificial Intelligence and Machine Learning

Special Awards

Harry and Laura Jacques Graduate Scholarship
Werner Graupe International Fellowship in Engineering

Arun Mishra Graduate Scholarship in Engineering

Reference Requests

Name	Email	Organization Name	Notified on
Mohamad, Mustafa	mustafa.mohamad@ucalgary.ca	University of Calgary	3 Nov 2023 6:41 PM
Armoush, Ashraf	armoush@najah.edu	An Najah National University	3 Nov 2023 6:41 PM

Academic History

Academic Institution	Degree/Diploma	From Date	To Date	Degree Date
Palestinian Territory-Occupie	Computer Engineering	2018/08/01	2022/12/31	2022/12/30
University of Calgary	MSc. ME Computational Science	2023/09/01	2025/07/01	2025/07/01

Scholarship History

Scholarship Name	Start Date	End Date	Annual Value	Funding Source	
Travel award	2022/11/17	2022/11/30	$1000.00	University	View Comment
Conference Award	2022/06/15	2022/07/02	$1500.00	University	View Comment
Research Fellow at Clemson uni	2021/10/01	2022/01/06	$3000.00	University	View Comment
Apple internship	2021/06/06	2021/09/19	$2000.00	Other	View Comment
The Hebba & Ishaq Sadaqah Foun	2020/09/11	2022/12/31	$3500.00	Provincial	View Comment
Faculty of Honor	2018/12/26	2021/09/30	$5000.00	University	View Comment
Community reputation Prize	2017/11/30	2023/01/26	$10000.00	Non-profit	View Comment
President of Palestine	2017/08/17	2022/12/31	$18000.00	Federal	View Comment

Research Proposal

***Title of research proposal (maximum 254 characters)**

Title: Particle Tracing and Predictive Machine Learning for Environmental Applications(Lagrangian Tracer Modeling)

***Abstract of research proposal (maximum 250 words)**

Abstract:

This research investigates the fusion of particle tracing techniques and predictive machine learning to advance environmental forecasting. The study aims to integrate computational particle tracing methods with sophisticated predictive algorithms to model particle behavior in fluid flows and forecast environmental conditions, particularly focusing on air pollution dispersion and weather patterns. The objectives encompass model development, air pollution forecasting enhancement, weather prediction improvement, and support for informed environmental decision-making. The methodology involves data collection, hybrid model development, training, validation, integration with weather models, continual evaluation, and visualization/reporting. Anticipated outcomes include enhanced air pollution forecasting aiding mitigation efforts, improved weather forecasts benefiting various sectors, and informed decision-making for environmental management. This research signifies the synergy between particle tracing and predictive machine learning, striving to offer precise insights for optimized environmental applications and public health protection.

***Research proposal (maximum 800 words)**

Title: Particle Tracing and Predictive Machine Learning for Environmental Applications Proposal

Introduction:
This proposal aims to integrate particle tracing techniques with predictive machine learning methods for advanced environmental applications. Particle tracing is a computational method used to model the movement of particles in fluid flows. By combining this technique with predictive machine learning, we can not only simulate particle behavior but also forecast future environmental conditions, such as air pollution dispersion and weather patterns. This proposal outlines the objectives, methodology, and potential outcomes of using particle tracing and predictive machine learning for environmental applications, including weather forecasting.

Objectives:

Develop a hybrid model: Integrate particle tracing techniques with predictive machine learning algorithms to create a comprehensive model capable of simulating particle behavior and predicting future environmental conditions.

Improve air pollution forecasting: Utilize historical air quality data, meteorological parameters, and particle tracing simulations to train machine learning algorithms for accurate and timely air pollution forecasting.

Enhance weather forecasting: Incorporate particle tracing simulations into weather prediction models to improve the accuracy and reliability of short-term and long-term weather forecasts.

Support environmental decision-making: Provide policymakers, environmental agencies, and stakeholders with valuable insights and predictions to aid in effective decision-making for pollution mitigation, resource management, and disaster response.

Methodology:

Data collection: Gather historical environmental data, including air quality measurements, meteorological data, particle tracking simulations, and other relevant information.

Model development: Develop a hybrid model that combines particle tracing simulations with machine learning algorithms, such as deep learning or ensemble methods, to predict particle behavior and environmental conditions.

Training and validation: Train the machine learning algorithms using historical data and validate the model's performance against observed data and known events.

Integration with weather models: Incorporate the particle tracing simulations and predictive machine learning algorithms into existing weather forecasting models to enhance the accuracy and reliability of predictions.

Evaluation and refinement: Continuously evaluate and refine the model's performance by comparing predictions with actual observations and incorporating new data to improve forecast accuracy.

Visualization and reporting: Present the predictions and insights in a visually compelling manner, utilizing graphs, maps, and other visualizations. Prepare comprehensive reports summarizing the research findings, methodologies employed, and potential applications.

Expected Outcomes:

Advanced air pollution forecasting: The integration of particle tracing and predictive machine learning will enable accurate and timely air pollution forecasts, aiding in pollution mitigation efforts and public health protection.

Improved weather forecasting: By incorporating particle tracing simulations, weather models can provide more precise and reliable forecasts, benefiting various sectors such as agriculture, transportation, and disaster preparedness.

Enhanced environmental decision-making: The research outcomes will provide valuable insights and predictions, supporting policymakers, environmental agencies, and stakeholders in making informed decisions related to pollution control, resource management, and disaster response.

Conclusion:
This proposal outlines the integration of particle tracing techniques with predictive machine learning for advanced environmental applications, including air pollution forecasting and weather prediction. By combining simulation and predictive modeling, this research aims to improve our understanding of particle behavior and enable accurate predictions of future environmental conditions. The proposed methodology will allow for the development of hybrid models that provide valuable insights and support decision-making for environmental management and public health protection.

Bibliography and Works Cited

*Bibliography and works cited (maximum 1000 words)

M. A. Mohamad, A. J. Majda, Eulerian and Lagrangian statistics in an exactly solvable turbulent shear model with a random background mean, Physics of Fluids, 2019.
M. A. Mohamad, A. J. Majda, Recovering the Eulerian energy spectrum from noisy Lagrangian tracers, Physica D, Jan 2020.
N. Chen, A. J. Majda, X. Tong, Information barriers for noisy Lagrangian tracers in filtering random incompressible flows, Nonlinearity, 2014.

A. J. Majda, P. Kramer, Simplified models for turbulent diffusion: theory, numerical modeling, and physical phenomena, Phys. Reports, 1999

Presentations and Contributions

Presentations and contributions (maximum 1000 words)

Presentation: Understanding the Lagrangian Filter for Estimating Fluid Velocity

Introduction

Imagine trying to understand the movement of fluid in a river or the air in our atmosphere. It's a complex task but essential for various fields like weather forecasting, environmental studies, and more. To estimate this movement, we often use innovative tools and techniques, one of which is the Lagrangian filter.

Understanding the Lagrangian Filter

The Lagrangian filter is a smart method used to estimate the speed and direction of fluid flow. It does this by observing the movement of tiny particles floating within the fluid, called tracer particles. These particles help us understand how the fluid moves by tracking their positions over time.

Key Components of the Lagrangian Filter

In our estimation process, we utilize specific notations and models. We look at the Eulerian velocity field, which describes the fluid's velocity at different points. Then there's the Lagrangian correlation function, which connects the fluid's velocity with the tracer particles' positions.

Filter Equations

To estimate the fluid's velocity accurately, we have mathematical equations that consider various factors like observation noise, dispersion, and random forcing terms. These equations, while complex, help us create estimates for the fluid's velocity at different locations and times.

Approximations for Practical Use

Calculating the exact solution using these equations can be challenging. Hence, we often use an approximate method called the Kalman-Bucy filter. This filter assumes certain properties about the correlation function, allowing us to make accurate estimations more easily.

Conclusion and Contributions

The Lagrangian filter, alongside its approximation, the Kalman-Bucy filter, plays a crucial role in helping us understand fluid dynamics. These tools assist scientists and researchers in estimating how fluids move, impacting various fields such as weather prediction, oceanography, and more. By using these innovative techniques, we gain valuable insights into the behavior of fluids, contributing significantly to scientific advancements and real-world applications.

Leadership Experience and Interpersonal Skills

***Leadership experience and interpersonal skills (maximum 800 words)**

I have an extensive background in computer science, software engineering, and research, demonstrating proficiency in various programming languages, frameworks, and tools. My educational journey at Najah National University showcases exemplary performance, ranking top in various technical and academic subjects. I've showcased versatility in software development, AI, data science, and machine learning across multiple roles and internships.
My professional experiences span several notable positions:
1. Harri , Machine Learning Engineer:

o Specialized in sales forecasting and time series analysis, working on predicting employee hiring needs for restaurants and retailers using real data.

2. CookOverflow (AI/ML Director and CTO, acquired by United Holy Land Fund):

o Transitioned the platform into an internal social media platform for UHLF alumni, focusing on professional referrals, connections, and communication.

3. Palestine Telecommunications Company (Paltel) (R&D Software Engineer - Backend):

o Contributed significantly to product development used by millions, engaging in the software development lifecycle and emphasizing collaboration and teamwork.

4. Apple Inc. (R&D Software Engineer for Apple TV):

o Gained expertise in Git, Python, databases, and backend frameworks (Flask, Django), and collaborated on projects involving clean code, database design, and backend concepts, project named Data stream interface.

5. Teaching and Research Roles:

o Assisted in Django/Python Bootcamps as a teaching assistant, fostering skills in data structures, object-oriented programming, and PostgreSQL.

o Engaged in undergraduate research at the University of Clemson, focusing on machine learning, data sets, Python, and high-performance computing, I lead a team of three researchers.

o TA for Arabs in Neuroscience organization funded by Simons Foundation.

o Volunteering teacher for Math undergraduate courses at NNU for first- and second-years students.
o Volunteering Swimming Teacher, I was in the Palestinian Olympic Team for swimmers in high school, and I was training children how to swim and select most talented for professional training.

6. Self R&D Projects and Achievements:

o Undertook numerous self-learning projects covering diverse areas like C++, Java, databases, web development, and Python (specifically in ML/AI, image processing, algorithms).

o Garnered several honors, scholarships, and awards, including best graduation project, scholarships, participation, and accolades in programming contests and technical certifications.
My strong motivation to pursue further education (MS/PhD) at the University of Calgary, evident in my detailed qualifications, project involvements, and research experience, highlights my commitment and passion for computational science, research, and learning.

I possess a multifaceted skill set, from technical proficiencies in multiple programming languages and frameworks to leadership, teaching, and volunteerism. My commitment to continuous learning and dedication to the field is evident in my extensive list of certifications and self-directed projects.

My commitment to advancing in the field, showcasing a genuine passion for technology, research, and problem-solving. My multifaceted experiences and consistent drive for self-improvement make me a promising candidate for advanced studies and impactful contributions to the field of computational science and technology.

Ahmad Droobi

UCalgary Profile: https://profiles.ucalgary.ca/ahmad-droobi
Linkedin : https://www.linkedin.com/in/droobi/

Allowable Inclusions

Allowable inclusions (maximum 250 words)

Applicant: Ahmad Droobi

Application Status: Submitted **Submission Date:** 13 Dec 2023 12:53 PM

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